Electronic Borehole Compass for Real-Time Borehole Trajectory Measurement and Borehole Survey

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1. Abstract

The present invention relates to an electronic borehole compass system designed for real-time measurement of borehole trajectory and surveying in geotechnical, exploratory, and blast-hole drilling operations in mining. Borehole deviation while drilling can cause serious problems such as ore dilution, high operational cost, poor fragmentation, and safety hazards. The proposed system comprises a combination of a magnetometer, an accelerometer, and a gyroscope chip to accurately find the orientation parameters, such as azimuth and inclination. The MCU acquires data from sensors such as accelerometers, gyroscopes, and magnetometers and combines them using a software-based technique known as sensor fusion to produce stable orientation and movement data. The technique is utilised to visualise the drill hole parameters.

Comparing it to conventional methods, the invention is a low-cost, space-saving, and high-precision solution with real-time feedback capability. It improves drilling accuracy, resulting in fewer deviation losses and better overall mining efficiency. The system is adaptable to a range of geological conditions and provides a scalable solution for today's technology-driven mining applications.

2. Literature Survey (Prior Art): (Please include patent and non-patent literature.)

2.1. Conventional Magnetic and Mechanical Surveying Instruments

Traditional single and multi-shot photographic borehole-surveying tools have been used for a long time in mineral exploration to determine borehole paths. Such systems usually have a **spherical compass**, where **the magnetic needle or sensing device is mounted inside a spherical (ball-shaped) housing** that allows it to rotate freely in all directions. This design enables the instrument to determine direction (or azimuth) even if it is tilted or non-horizontal.

These conventional instruments are functional, but they have limitations. They are sensitive to magnetic interference from drill strings, casing, and magnetic ore bodies, such as magnetite. Besides, the use of these tools often requires the drilling process to be stopped so that discrete point measurements can be taken. This greatly reduces the operational efficiency. They also tend to use analog readouts or delayed post-processing instead of real-time digital data. Thus, it is not always possible to adjust the trajectory in real time while performing regular mining activities.

2.2. Gyroscopic and Advanced Inertial Measurement Systems

To overcome magnetic interference, the industry uses various gyroscopic surveying tools, such as mechanical rate gyroscopes and Fibre Optic Gyroscopes (FOG). These tools often work with Inertial Navigation Systems (INS) and are widely used in the deep-well oil & gas sector to deliver continuous, high-accuracy, three-dimensional borehole profiles.

However, these commercial systems are sensitive and complex and come with significant capital and operating costs. They also require precise calibration methods and highly trained personnel. This makes them too complicated and expensive for typical, high-volume, cost-sensitive mining exploration and blast-hole monitoring. There is a critical need for a compact, tough, and affordable digital borehole compass designed for the quick deployment needs of the mining engineering sector.

2.3. Notable prior art of the Invention

Method and Apparatus for Determining Inclination Angle of a Borehole While Drilling (US 5,128,867, 1992) uses downhole magnetometers and accelerometers to figure out the borehole azimuth and inclination while the drill string is rotating. It provides a method to combine readings of the total magnetic field with dip angle for directional tracking. However, the system is extremely vulnerable to localised magnetic anomalies, quite common in metalliferous mining sites, mainly because the magnetic sensors are left unshielded and that makes them pick up unwanted interference. Also, this approach does not provide a methodology for real time IoT integration or for continuous three dimensional trajectory visualisation at the surface.

Borehole Survey System Utilising Strapdown Inertial Navigation (US 4,812,977, 1989) combines strapdown accelerometers and angular rate sensors to keep mapping the borehole path through the whole wireline deployment. It represents an early approach of using inertial information for surveying. However, this legacy system relies on outdated sensor processing methods and wireline velocity assistance. Also, there is no modern MEMS (Micro-Electro-Mechanical Systems) miniaturisation, specialised shock resistance dampening, and automated calibration algorithms, needed to survive the heavy vibration that's common in mining. It also lacks a synchronised data transmission architecture that would fit digital mine management.

The Multi-Gimbaled Borehole Navigation System (US 7,093,370, 2006) shows an omnidirectional kind of navigation setup that uses solid-state gyros and accelerometers stacked inside inner gimbals. Despite having a compact form factor for smaller-diameter drill pipes, it still relies on a very complicated and delicate mechanical gimbaling mechanism instead of using a completely solid-state strapdown inertial measurement unit. This mechanical complexity makes the tool economically unfeasible and more likely to fail under the high-impact conditions of standard mining operations, unlike the ruggedised, cost-effective, and fully solid-state architecture required for routine mining deployment.

3. Detailed Description of the Invention, including drawings:

This is the most important section. Describe your invention as clearly and in as much detail as possible. You do not need to provide technical terms or legal language. Focus on explaining what it does, how it works, and its main components.

Key Areas to Address:

Purpose of the Invention: What problem does it solve?

Main Components: What are the key parts? (e.g., parts of a device, software modules, etc.)

How It Works: Describe the step-by-step process of how the invention functions.

Benefits: Why is your invention better or different from existing products or methods?

3.1 Purpose of the Invention

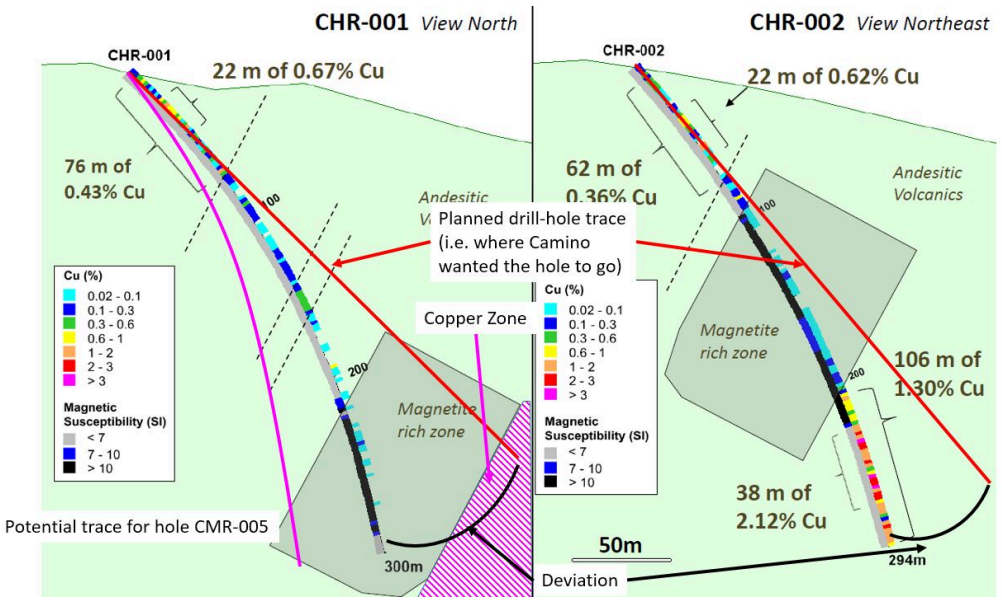
The main purpose of the disclosed invention is to improve the accuracy and reliability of trajectory measurement and orientation of boreholes in mining and related areas. Proper positioning of boreholes plays a crucial role in obtaining dependable geological information during exploration for better estimation of ore body boundaries, ore thickness and the grade distribution. In this way, resource modeling gets improved, and the uncertainties in mine planning get reduced too.

The invention also aims to reduce deviations in blast hole drilling by offering accurate monitoring and direction control of the borehole. This ensures proper burden, spacing, and alignment of blast holes, resulting in uniform energy distribution during blasting, controlled fragmentation, and reduction of hazards such as flyrock, excessive vibrations, and unblasted zones.

The invention also seeks to increase the accuracy of fracture orientation and delineation in stress measurement techniques such as hydraulic fracturing. Accurate data of borehole geometry are required for reliable interpretation of induced fractures and in-situ stress conditions, as well as for improved identification of structural features such as joints and faults. Hence, it can be said that the invention improves the accuracy of the borehole data and therefore enables safer, more efficient, and optimised mining operations.

3.1.1 Borehole trajectory measurement to improve resource modelling and reduce uncertainties in mine planning

The figure below (camino-minerals-blogspot) is a good example of how borehole deviation can affect ore intersection and interpretation.



The **planned drill path** in both drill holes (CHR-001 and CHR-002), shown by the straight red line, is the correct intersection path to intersect the copper zone. But the **actual borehole path** does not follow the plan, as can be seen from the bent trace. This deflection causes the borehole to intersect the ore body at a different location and orientation.

As a consequence of this misalignment:

- The borehole may **intersect the ore body at an angle**, which can make the apparent thickness of mineralisation appear bigger or smaller. For example, an ore zone may appear thicker than it actually is due to a shallow intersection.
- It may miss high-grade zones (red/pink areas in legend) or hit them partially, causing an under-estimation of grade.
- The borehole may also intersect adjacent zones (e.g., magnetite-rich zones), giving misleading information not representative of the true ore geometry.

This deviation results in a long intersection (106 m at 1.30% Cu) as seen in CHR-002, but this may not be representative of the true vertical thickness of the ore body due to the angulated path of the borehole.

Thus, proper alignment of the borehole is required to:

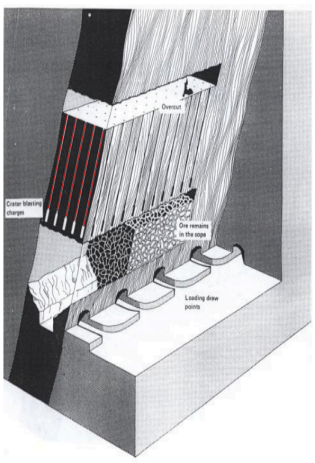
- Intersect the ore body at the right position and orientation
- Deliver true thickness measurements
- Reflects the actual grade distribution

It is evident from the figure that even small deviations can cause notable distortions in geological interpretation, leading to mistakes in resource estimation and mine planning.

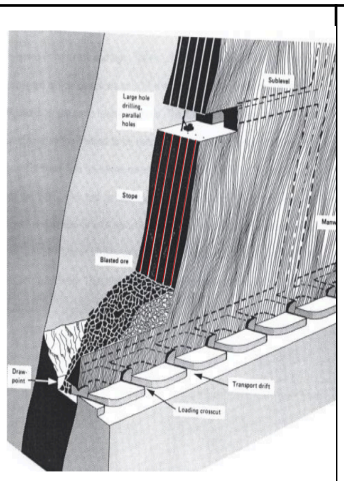
3.1.2 Minimising deviation in blast hole drilling through continuous monitoring and control of the borehole direction

Accurate control and monitoring of borehole direction are crucial for reducing deviations in blast hole drilling, mainly when operations are done with long-hole mining methods such as

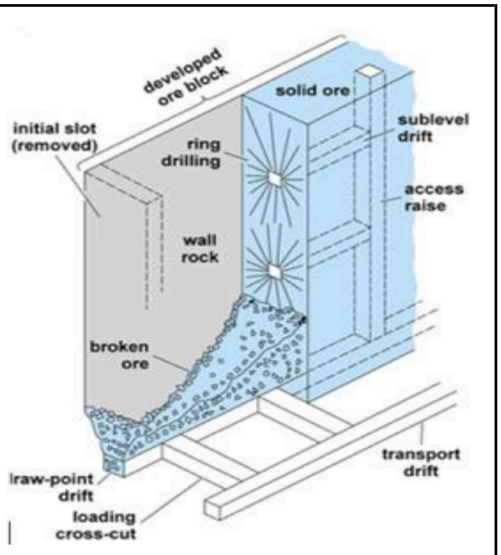
sublevel stoping and Vertical Crater Retreat (VCR). If borehole deviation is not handled properly, it turns into large operational inefficiencies, such as ore dilution, ore loss, poor fragmentation, and higher operating costs. Hence, there is a need to work on robust monitoring and control strategies to keep the drilling precision in check.



Sublevel open stoping method



VCR Mining Method



Sub-level open stoping

A borehole deviation report becomes a simple but important instrument for reaching this goal. It gives a quantitative look at drilling accuracy through structured recording and evaluation of deviations across factors such as hole trajectory, burden, spacing, and depth. With this, engineers can identify the usual deviations, then apply corrective measures in the next drilling cycles. In the end, it improves both control and predictability of drilling operations, and that also helps blast designs come out more accurate.

The reports are built around the use of downhole monitoring devices, like borehole compasses and deviation measuring tools. These tools give live feedback or post-drilling review of the hole paths, and also flag early signs if the well is drifting away from the planned design. Including this information in a deviation report helps in maintaining the optimum burden and spacing, which is critical for a consistent energy spread during the blasting stage. Better rock breakage comes from that proper energy distribution, so in the end, there are fewer oversized boulders, and there is less need for any later, secondary blasting.

Also, borehole deviation reports help with better utilisation of the equipment and more practical maintenance planning. These deviation trends can be analysed to check if the deviations come from equipment-related causes, such as drill string instability, worn-out bits, or misaligned rigs. Addressing these problems results in more accurate drilling, better operational efficiency, and fewer mechanical failures.

Deviation reporting also helps optimise drilling parameters. Engineers can figure out the best drilling conditions for specific geological formations by looking at deviation data against

parameters like thrust, rotation and flushing velocity. As a result, the trajectory is deviated less, especially in heterogeneous or anisotropic rock masses, and the blasting results are, therefore improved.

Geological variability is another major factor that affects borehole deviation. The integration of geological data and deviation reports gives a better understanding of how rock properties like layer orientation, joint orientation, and hardness influence the drilling behaviour. Then, predictive adjustments can be made to the drilling design so that boreholes stay on course even when the geology gets complicated or unpredictable.

Borehole deviation reports are also useful for training operators and evaluating their performance. When human errors are detected in the alignment and execution of drilling, targeted training programs become possible, which in turn boosts operator accuracy. This makes drilling practices more consistent across different shifts and different people.

At a broader operational level, combining deviation reports with computer-aided tools for blast design and simulation pushes mining efficiency up. Accurate deviation data improves blast modelling, which ultimately results in better rock fragmentation, reduced energy consumption during crushing and grinding, and smoother downstream mineral processing operations.

In conclusion, reporting of borehole deviation is vital to improve mining efficiency and blast fragmentation to allow data-driven decisions. This helps in accurate drilling, optimising blast design, minimising operational losses and improving overall productivity. Reducing deviations results in better fragmentation, lower costs and safer mining operations. Deviation reporting is therefore an essential part of modern mining practices.

3.1.3 Fracture orientation and delineation in stress measurement techniques such as hydraulic fracturing

The determination of the subsurface state of stress depends critically on the orientation of the fractures that develop on the borehole walls. The hydraulic fracture orientation indicates the direction of the stress field because inclined and vertical fractures grow perpendicular to the minimum horizontal principal stress. While creating a fracture, there is a significant increase in tangential tension in the borehole before fracturing, and there is a minimal effect on the stresses along the borehole axis. Hydraulic fractures tend to initiate in a plane that passes through the borehole axis, even when the minimum principal stress is vertical or when the rock contains significant strength anisotropy. Generally, the most common fracture orientations are vertical or steeply dipping along the borehole, unless horizontal stresses are significant relative to vertical stresses or there is extreme strength anisotropy (bedding or foliation). Typically, impression packers are used to determine fracture orientation. The impression of the induced vertical hydraulic fractures on impression packers is rarely found in the form of two straight lines 180 degrees apart.

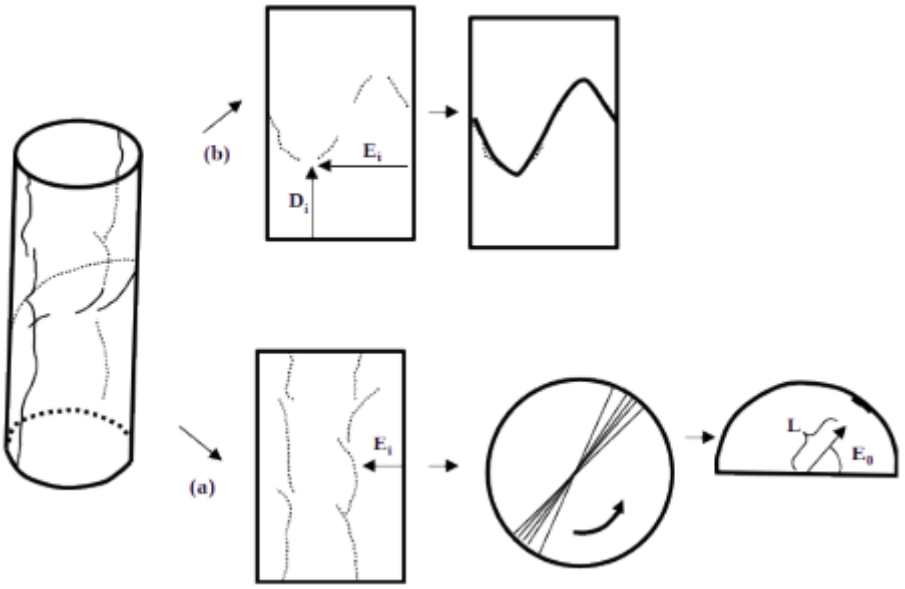


Figure 1: Fracture delineation from an impression packer. (a) Circular statistics applied to sets of vertical fractures, (b) sinusoidal regression Applied to an inclined fracture

3.2 Parts of a device, software modules, etc.

The invention consists of an electromechanical system that includes a durable mechanical enclosure and a specially created printed circuit board (PCB) together with two software layers that consist of embedded firmware and external data-processing software.

3.2.1 Electronic Circuitry and Hardware

The custom PCB enables sensing and logging functions because it securely connects tiny solid-state sensors. The key functional components include: Microcontroller Unit (MCU):

- A central processing unit that operates continuously, gathering data while it processes multiple sensors and performs tilt correction. The multi-axis accelerometer detects static acceleration forces across three axes via its solid-state inertial sensor, which uses a PCB system.
- The component identifies the device's spatial orientation through its measurement of both pitch and roll angles as they relate to gravitational forces. The solid-state magnetometer functions as a multi-axis digital compass system that measures local magnetic field strength through its integrated circuitry.
- The system determines horizontal direction (yaw/azimuth) through its calculation of directional orientation towards magnetic north.
- The Non-Volatile Data Storage Module acts as a secure memory interface that uses standard serial bus connections to store time-stamped sensor data in both raw and processed formats.
- The Power Management and Communication Interfaces system uses its onboard voltage regulation to maintain stable logic signal levels, while existing power supply

and standard serial communication systems enable external programming and data extraction functions.

3.2.2 Mechanical Housing and Safety Features

- The pressure-resistant casing protects internal electronics by sealing them inside a durable cylindrical housing that can endure extreme hydrostatic pressures and downhole abrasive conditions.
- The Integrated Shock Absorption System uses a mechanical dampening mechanism inside its casing assembly, which protects the sensitive PCB from harmful high-frequency impact vibrations that occur during drilling.

3.2.3 Software Stack (Embedded and Post-Processing)

- Embedded Data Acquisition & Fusion Firmware: The embedded firmware samples raw sensor data continuously at a fixed rate (for instance, every second) to generate trend and plunge measurements.
- Post-Processing Software Suite: The software suite uses logged data as input, which it processes by applying digital signal filtering methods to eliminate sensor noise, and it performs coordinate transformation methods to change spherical data into 3D Cartesian coordinates.

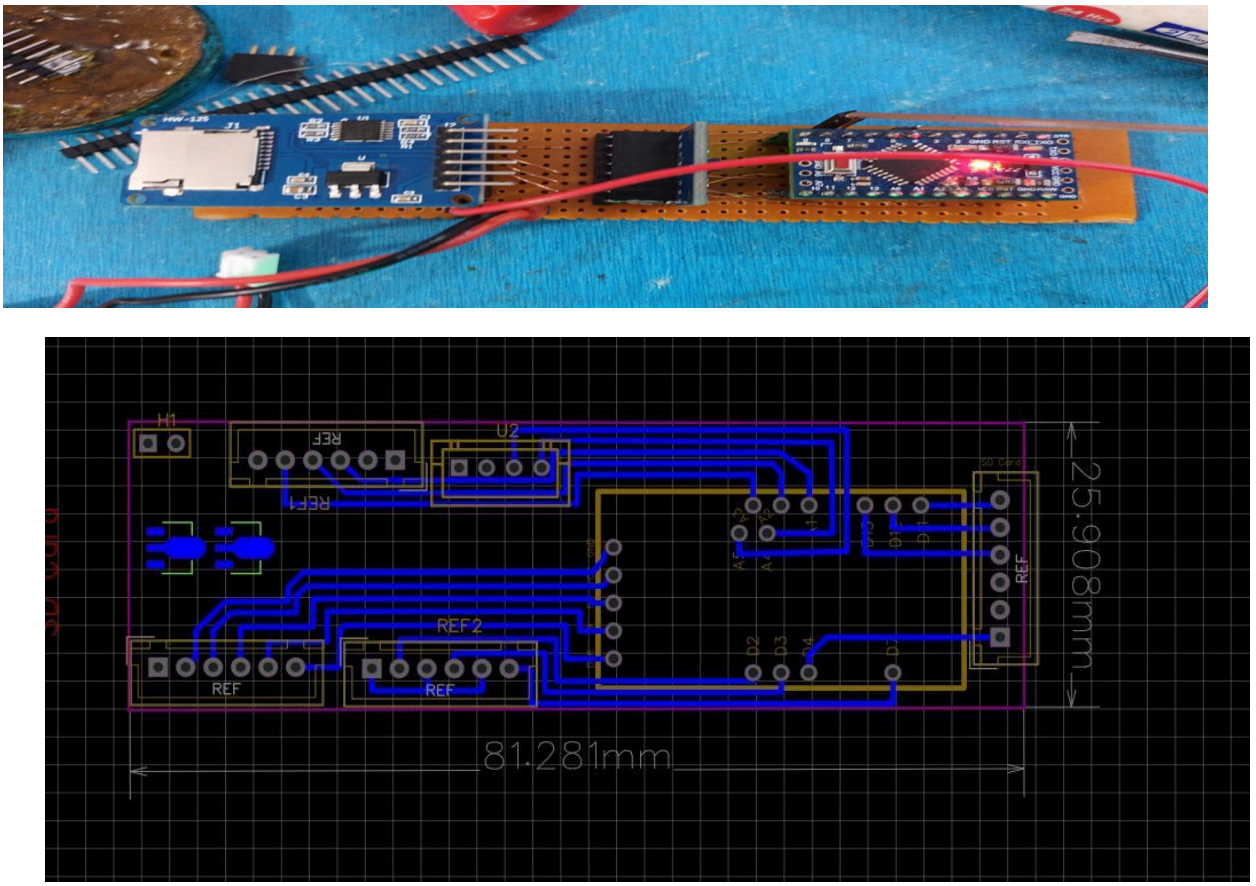


Fig.: designed PCB incorporating all the sensors

3.2.4 Mobile Application and Wireless Communication System

The invention further includes a dedicated mobile application especially designed for mining and geotechnical engineering. Its primary purpose is to map the exact 3D underground trajectory of a drilled borehole. It pairs wirelessly with a hardware probe lowered into the earth, processes raw gravitational and magnetic data, and visualizes the hole's path in real time.

Features of the Mobile Application:

- **Bluetooth & Hardware Integration**
  The application uses Bluetooth Serial Port Profile (SPP) communication to scan for and connect with the borehole surveying probe. Once connected, the system maintains continuous real-time communication with the device and receives high-speed survey data packets.
  It also uses a buffering mechanism to make sure that no data is lost even if Bluetooth packets are fragmented during transmission. Once the device is calibrated, the application is able to detect the connected probe and then, it loads its unique 46 point calibration matrix from local storage to maximize accuracy.

- **Live Survey Operations**
  The software shows a “New Survey” option, which helps operators start a fresh borehole survey by giving it a distinct Project ID, and then typing the starting depth where the drill bit is positioned.
  The survey can run in two different modes: **(1) Manual Mode**, where the operator presses a button to take one reading at a time, while the app makes you wait through a stabilization countdown, so the probe is totally steady before recording;
  (2)Auto-Log Mode, where the app runs on a continuous timer (for example every 30 seconds), and it keeps recording automatically and moving the depth forward by a set interval, say 3 meters or so.
  During a survey, the application continuously displays live borehole metrics on-screen. This includes depth, azimuth, inclination (dip/zenith), trend and plunge, and also the gyroscope readings as they come in, in real time.
  The software automatically puts a time stamp on the data it gets and saves it so people can look at it later and figure out the path the borehole took.

- **Mathematical Processing and Orientation Calculation**
  The app collects raw Accelerometer (gravity) and Magnetometer (Magnetic North) readings from the probe. Then, during calibration, it uses the hardware specific hard-iron offsets and 3x3 soft-iron correction matrices, to get rid of magnetic interference, sensor offset and alignment errors (It’s like aligning everything so it behaves right).
  After that, with tilt-compensation and some trigonometric processing algorithms, the software calculates azimuth (basically heading direction), inclination or zenith angle (how much the borehole tilts away from vertical), and roll or toolface angle (the rotational angle of the probe inside the pipe). All these computations give reliable

borehole orientation and directional drilling data, in real time, without the usual guesswork.

- **3D Borehole Trajectory Reconstruction**
  As the device collects data points, the app rebuilds the borehole path using the Balanced Tangential Method, which is an industry-standard approach. Then the software changes the survey angles and depth measurements into 3D spatial coordinates: North (Y), East (X), and True Vertical Depth (Z). This can help visualise the actual path taken by the borehole under the ground more accurately.

- **Offline Deep-Hole Synchronization**
  In deep underground operations where Bluetooth signals cannot really penetrate the surrounding rock strata, the probe keeps on recording survey data and storing it away in onboard memory.
  Later, when the probe is pulled back up to the surface and Bluetooth reconnects, the operator can use the Sync Offline Data option to grab everything that was saved. The app then pushes a command ($DUMP*) into the probe, downloads every offline shot, processes it, and brings it back into the present 3D trajectory, seamlessly. This feature enables uninterrupted data collection during deep-hole surveying work, even when the link is basically missing.

- **Utility & Quality Control Tools**
  The application also features engineering utility and diagnostic tools, so live orientation can be monitored, gravity vector checks can happen, angle conversions are handled, and survey validation runs more smoothly.
  The gravity vector verification tool basically monitors the overall gravity vector that comes from the accelerometer system. When the measured gravity somehow strays a lot from the usual reference of 1.0 g, because of excessive vibration, unstable probe position, or maybe a hardware malfunction, the application then raises an “OUT OF RANGE” warning automatically, to warn the operator. This feature ensures stability of measurements, improves survey reliability, and reduces inaccurate borehole readings during drilling operations.

- **Data Management & Export**
  All survey data, including raw sensor readings, processed orientation data, and calculated coordinates, are stored permanently in an embedded NoSQL database (Isar) on the mobile device.
  The application also lets users pull up earlier borehole projects, sort out, filter and check past survey records, and export the full survey dataset into spreadsheet layouts, like Excel (.xlsx). This kind of approach helps with documentation, reporting, and integration with mine planning workflows, a little more smoothly.

- **System Security and Data Integrity**
  The application is built to work mostly offline, so the sensitive geological and mining information stays protected. It keeps survey data and calibration records stored locally in the device database, so it reduces any unnecessary upload or transfer of underground survey material to outside servers.

Also, the software is set to run only with authorised and calibrated borehole probe hardware, to keep measurement reliability and overall system accuracy in check. Finally, the application includes operational safety instructions and user acknowledgment screens, for correct field deployment.

- **Integration with the Borehole Compass System**
  The mobile application works like an integrated part of the electronic borehole compass system. The overall working process includes:
  1. Calibration and initialisation of the borehole probe;
  2. Bluetooth pairing between the probe and mobile application;
  3. Continuous acquisition of accelerometer, gyroscope, and magnetometer data;
  4. Real time display plus logging of the borehole orientation;
  5. Local storage, and then synchronization of survey data when possible;
  6. Three dimensional borehole trajectory reconstruction, and finally export.

  So, when you put together the MEMS sensors, the embedded electronics, and the wireless mobile app, you get a compact, low-cost and user-friendly borehole surveying setup, fit for modern mining operations.

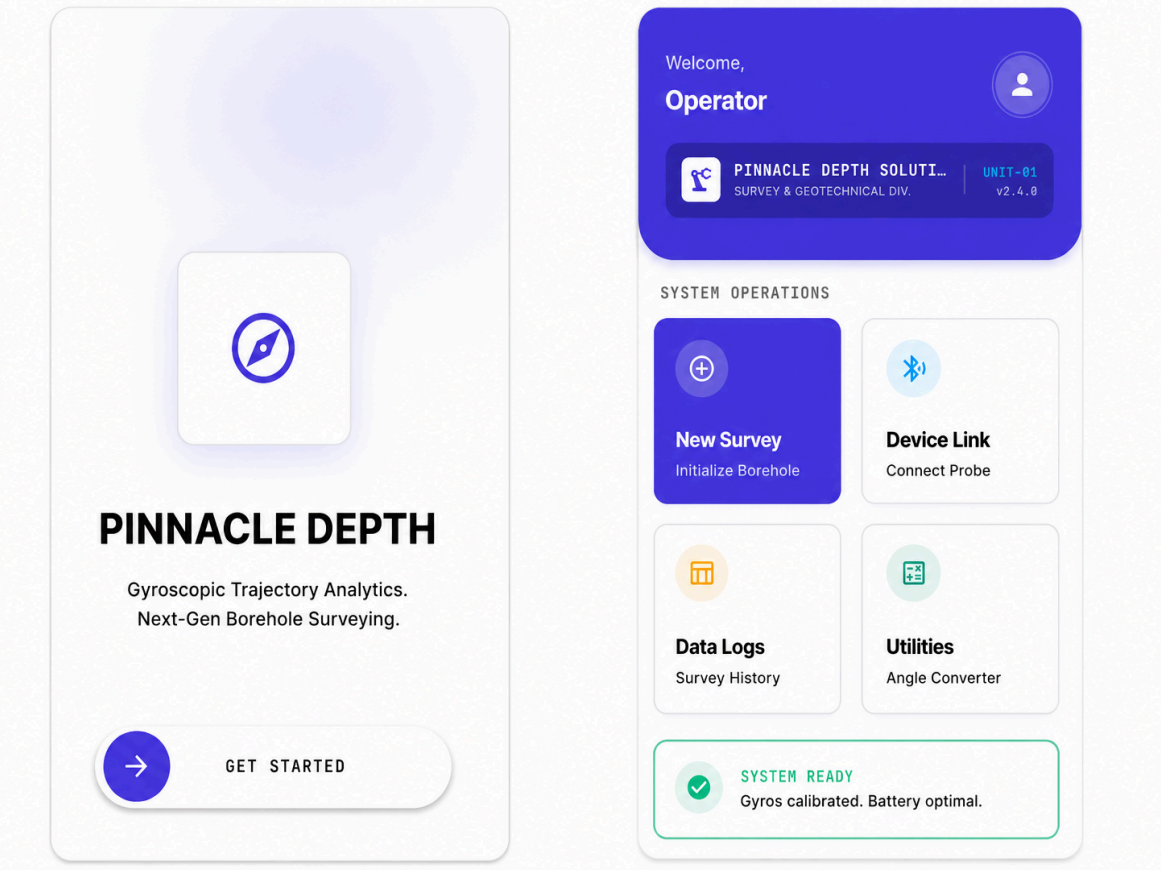


Fig.: Application Interface of the Electronic Borehole Compass

3.3 How It Works: Describe the step-by-step process of how the invention functions.

The invention operates through a sequence of stages that demonstrate its complete functioning. The electronic borehole compass transforms actual underground movements into a full three-dimensional digital path mapping system through these operational steps.

1. **Surface Vector Calibration:** The device needs to complete physical calibration at the surface before its usage in downhole operations. The Total Station surveying equipment enables the measurement of Cartesian coordinates for two different physical points, which are located on the linear mechanical housing. The centre point and the extreme top end of the housing serve as the two measurement points. The measured physical vector is mathematically converted into spherical coordinates, which are known as trend and plunge. The true value is compared with the trend and plunge data that the sensor recorded. The resulting difference creates the error-correction offsets, which need to be computed.
2. **Deployment and Continuous Data Acquisition:**
   The drill string receives the secure installation of the calibrated casing. The onboard MCU begins its sensor initialisation process when the assembly moves into the borehole. The accelerometer and magnetometer record their respective acceleration forces and magnetic field strength. The mechanical shock absorption system reduces drill vibrations.
3. **Real-Time Signal Processing and Spatial Coordinate Transformation:** The onboard microcontroller handles the processing of all incoming sensor data. The firmware uses gravitational vectors from the accelerometer to create a magnetic levelling system based on magnetometer measurements. The tilt-compensation algorithm converts all incoming sensor data into standard geological coordinates, which include trend and plunge measurements.
4. **Autonomous High-Frequency Data Logging:** The computed orientation parameters are time-stamped and written immediately to the onboard non-volatile storage module at fixed frequency intervals. This localised storage system works as a backup arrangement that maintains data integrity when deep underground signal dropouts occur.
5. **Data Retrieval, Digital Filtering, and 3D Trajectory Reconstruction:** The device recovers its localised data file from the system. The time series orientation dataset lines up with the depth measurements, which the operator records. Special external post-processing software applies digital signal filtering to suppress mechanical and magnetic noise irregularities. After that, the software carries out a mathematical transformation that changes the corrected spherical coordinates into actual 3D Cartesian coordinates. Finally, the resultant Cartesian data is plotted graphically, which enables the engineers to visually assess the real spatial geometry and the deviation pattern of the borehole.

3.4 Benefits: Why is your invention better or different from existing products or methods?

The electronic borehole compass offers several useful advantages in mining and geotechnical applications, as discussed below:

1. **Accurate Borehole Deviation Measurement:**
   With this invention, it becomes possible to determine the azimuth and borehole inclination very precisely while drilling is still underway. That helps spot any divergence from the intended borehole line early, so the need for re-drilling drops, and ore loss is also minimised. This, in turn, boosts the general efficiency of the workflow.
2. **Wireless Connectivity (Bluetooth Integration):**
   The device relies on Bluetooth-based wireless communication so data can be moved quickly from the tool to external devices for monitoring, storage, and later analysis, basically without hassle.
3. **Directional Drilling Support:**
   It gives real-time information about borehole positioning and orientation, which supports operator control during directional drilling. This improves drilling precision for both exploration and development, and it also helps in maintaining the right drilling path even when the geological formations start to change.
4. **In-situ Stress Orientation Analysis:**
   This device can be used to support hydrofracturing stress measurements by figuring out the directional spread of the fractures that get induced. The in situ stress directions in underground rock masses can be inferred from the measured orientation data coming from the boreholes. This kind of information is very crucial for the design of mine openings, pillars, and other underground excavations, and it leads to better overall structural stability.
5. **Improved Mine Planning and Safety:**
   The detailed borehole deviation information and stress orientation data, which are obtained through the disclosed system, enable more solid mine planning. It also supports ground condition evaluation, and it improves how underground excavations are designed. This results in better stability of the ground and more safety during operations.
6. **Economic Feasibility:**
   Conventional borehole deviation and in-situ stress measurement tools, which are used in the oil and gas and mining industries are bulky, complex, and expensive, making them unsuitable for regular use in underground mining. This becomes even harder in cost-sensitive settings like Indian mines.
   The disclosure here presents a compact borehole logging arrangement with Micro Electro Mechanical Systems, or MEMS, sensors that digitally capture borehole deviation and in situ stress parameters. By using commercially available, low power, smartphone grade inertial and orientation sensors, the system lowers manufacturing complexity and component costs, along with power consumption while still keeping measurements accurate and reliable.
7. **Advanced Field-Tested Applications:**
   The device has been successfully deployed in underground mining environments (for example, Moonidih and Churcha mines) and is found to be able to measure the orientation of hydrofracture stresses even at depths more than 200 m. This capability makes it possible to analyse in situ stress fields and provide useful input data for geotechnical analysis, stability evaluation, and mine planning.



Figure 3.4.1: Field Testing of the device at Churcha Mines (SECL), showing its work ability in deep underground mining situations.



Figure 3.4.2: Testing team comprising Prof. RK Sinha and the student team
(Location – XV Seam Monorail Incline, Moonidih Mines)



Figure 3.4.3: Team after successful testing of Electronic Borehole Compass
(Location- XV Seam, 7D/7L at 635m depth, Moonidih Mines)